

DEPARTMENT OF CSE(IOT)

Question Bank for Comprehensive Engineering Aptitude Test

Multiple Choice Questions (1 Marks)

1. The amount of ROM needed to implement a 4 bit multiplier is

- A. 64 bits
- B. 128 bits
- C. 1 Kbits
- D. 2 Kbits

2. A computer has a 256 KB, 4-way set associative, write back data cache with block size of 32 bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.

The number of bits in the tag field of an address is-

- A. 11
- B. 14
- C. 16
- D. 27

3. Consider the following sequence of micro-operations.

$MBR \leftarrow PC$

$MAR \leftarrow X$

$PC \leftarrow Y$

$Memory \leftarrow MBR$

Which one of the following is a possible operation performed by this sequence?

- A. Instruction fetch
- B. Operand fetch
- C. Conditional branch
- D. Initiation of interrupt service

4. The following code segment is executed on a processor which allows only register operands in its instructions. Each instruction can have at most two source operands and one destination operand. Assume that all variables are dead after this code segment.

```
c = a + b;  
d = c * a;  
e = c + a;  
x = c * c;  
if (x > a) {  
    y = a * a;  
}  
else {
```

```
 $d = d * d;$ 
```

```
 $e = e * e;$ 
```

```
}
```

Suppose the instruction set architecture of the processor has only two registers. The only allowed compiler optimization is code motion, which moves statements from one place to another while preserving correctness. What is the minimum number of spills to memory in the compiled code?

- A. 0
- B. 1
- C. 2
- D. 3

5. In a k -way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set $(s+1)$. The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from.

- A. $(j \bmod v) * k$ to $(j \bmod v) * k + (k-1)$
- B. $(j \bmod v)$ to $(j \bmod v) + (k-1)$
- C. $(j \bmod k)$ to $(j \bmod k) + (v-1)$
- D. $(j \bmod k) * v$ to $(j \bmod k) * v + (v-1)$

6. Which operating system is commonly used with Raspberry Pi for IoT projects?

- A. Windows XP
- B. Ubuntu Mobile
- C. Raspberry Pi OS
- D. MS-DOS

7. The smallest integer that can be represented by an 8-bit number in 2's complement form is

- A. -256
- B. -128
- C. -127
- D. 0

8. Which one of the following is the tightest upper bound that represents the time complexity of inserting an object into a binary search tree of n nodes?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \log n)$

9. Which one of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

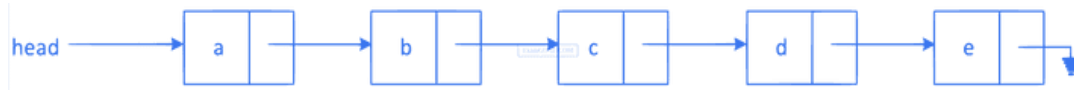
- A. $O(\log n)$
- B. $O(n)$
- C. $O(n \log n)$

D. $O(n^2)$

10. Consider a hypothetical processor with an instruction of type $LW\ R1, 20(R2)$, which during execution reads a 32-bit word from memory and stores it in a 32-bit register $R1$. The effective address of the memory location is obtained by the addition of a constant 20 and the contents of register $R2$. Which of the following best reflects the addressing mode implemented by this instruction for operand in memory?

- A. Immediate Addressing
- B. Register Addressing
- C. Register Indirect Scaled Addressing
- D. DBase Indexed Addressing

11. Consider the problem of reversing a singly linked list. To take an example, given the linked list below:



the reversed linked list should look like



Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in $O(1)$ space?

- A. The best algorithm for the problem takes $\Theta(n)$ time in the worst case.
- B. The best algorithm for the problem takes $\Theta(n \log n)$ time in the worst case.
- C. The best algorithm for the problem takes $\Theta(n^2)$ time in the worst case.
- D. It is not possible to reverse a singly linked list in $O(1)$ space.

11. The result evaluating the postfix expression $10\ 5+60\ 6/*8-$ is

- A. 284
- B. 213
- C. 142
- D. 71

12. Which connectivity option is commonly available in modern **Raspberry Pi** boards for IoT communication?

- A. Wi-Fi
- B. Bluetooth
- C. Ethernet
- D. All of the above

14. A function f defined on stacks of integers satisfies the following properties. $f(\varphi) = 0$ and $f(\text{push}(S, i)) = \max(f(S), 0) + i$ for all stacks S and integers i . If a stack S contains the integers 2, -3, 2,

-1, 2 in order from bottom to top, what is f(S)? A. 6 B. 4 C. 3 D. 2
15.Consider the tree arcs of a BFS traversal from a source node W in an unweighted, connected, undirected graph. The tree T formed by the tree arcs is a data structure for computing A. The shortest path between every pair of vertices. B. The shortest path from W to every vertex in the graph. C. The shortest path from W to only those nodes that are leaves of T. D. The longest path in the graph.
16.Consider the following C program segment. <pre>#include <stdio.h > int main() { char s1[7] = "1234", *p; p = s1 + 2; *p = '0'; printf("%s", s1); }</pre> What will be printed by the program? A. 12. B. 120400. C. 1204. D. 1034.
17.Raspbian is mainly stored and booted from: A. Hard Disk B. USB Drive C. SD Card D. CD-ROM
18.In Python GPIO programming, which function is used to set the pin mode? A. GPIO.start() B. GPIO.mode() C. GPIO.setmode() D. GPIO.begin()
19.Consider the following array of elements. {89,19,50,17,12,15,2,5,7,11,6,9,100} The minimum number of interchanges needed to convert it into a max-heap is A. 4. B. 5.

C. 2. D. 3.
20. Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE? P: Minimum spanning tree of G does not change Q: Shortest path between any pair of vertices does not change A. P Only. B. Q Only. C. Neither P nor Q. D. Both P and Q.
21. To implement Dijkstra's shortest path algorithm on unweighted graphs so that it runs in linear time, the data structure to be used is: A. Queue. B. Stack. C. Heap. D. B-Tree.
22. The Floyd-Warshall algorithm for all-pair shortest paths computation is based on A. Greedy paradigm. B. Divide- and- conquer paradigm. C. Dynamic programming paradigm. D. Neither Greedy nor Divide- and- conquer nor Dynamic programming paradigm.
23. Match the following: List 1 (P) Prim's algorithm for minimum spanning tree (Q) Floyd-Warshall algorithm for all pairs shortest paths (R) Mergesort (S) Hamiltonian circuit List 2 (i) Backtracking (ii) Greedy method (iii) Dynamic programming (iv) Divide and conquer A. P-iii, Q-ii, R-iv, S-i. B. P-i, Q-ii, R-iv, S-iii. C. P-ii, Q-iii, R-iv, S-i. D. P-ii, Q-i, R-iii, S-iv.
24. Let S be an NP-complete problem and Q and R be two other problems not known to be in NP. Q is polynomial time reducible to S and S is polynomial-time reducible to R. Which one of the following statements is true? A. R is NP-complete.

- B. R is NP-hard.
- C. Q is NP-complete.
- D. Q is NP-hard.

Multiple Choice Questions (2 Marks)

1. Consider a 4-way set associative mapping with 16 cache blocks. The memory block requests are in the order-
0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155
If LRU replacement policy is used, which cache block will not be present in the cache?
A. 3
B. 8
C. 129
D. 216

2. On a non-pipelined sequential processor, a program segment, which is a part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

Initialize the address register

Initialize the count to 500

LOOP: Load a byte from device

Store in memory at address given by address register

Increment the address register

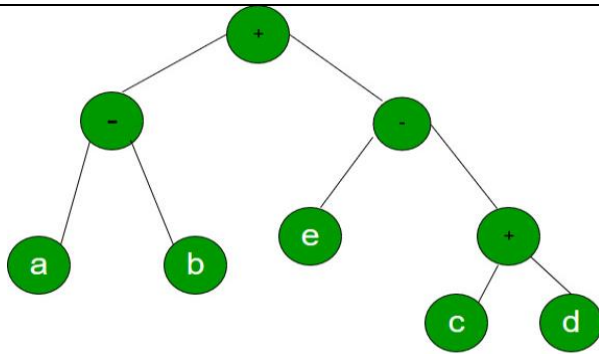
Decrement the count

If count $\neq$ 0 go to LOOP

Assume that each statement in this program is equivalent to machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute. The designer of the system also has an alternate approach of using DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory. What is the approximate speedup when the DMA controller based design is used in place of the interrupt driven program based input-output?

- A. 3.4
- B. 4.4
- C. 5.1
- D. 6.7

3. Consider evaluating the following expression tree on a machine with load-store architecture in which memory can be accessed only through load and store instructions. The variables a, b, c, d and e initially stored in memory. The binary operators used in this expression tree can be evaluating by the machine only when the operands are in registers. The instructions produce results only in a register. If no intermediate results can be stored in memory, what is the minimum number of registers needed to evaluate this expression?



- A. 2
- B. 9
- C. 5
- D. 3

4. A RAM chip has a capacity of 1024 words of 8 bits each (1K * 8). The number of 2 * 4 decoders with enable line needed to construct a 16K * 16 RAM from 1K * 8 RAM is

- (A) 4 (B) 5 (C) 6 (D) 7

5. Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions I1, I2, I3, ..., I12 is executed in this pipelined processor. Instruction I4 is the only branch instruction and its branch target is I9. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

- A. 132
- B. 165
- C. 176
- D. 328

6. Which function is used to set the pin numbering mode?

- A. `GPIO.setmode()`
- B. `GPIO.setup()`
- C. `GPIO.init()`
- D. `GPIO.pinmode()`

6. The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, and 42. Which one of the following is the post order traversal sequence of the same tree?

- A. 10, 20, 15, 23, 25, 35, 42, 39, 30
- B. 15, 10, 25, 23, 20, 42, 35, 39, 30
- C. 15, 20, 10, 23, 25, 42, 35, 39, 30
- D. 15, 10, 23, 25, 20, 35, 42, 39, 30

7.How do you create a button in Tkinter?

- A. `Button(master, text="Click Me", command=func)`
- B. `Button.create(master, "Click Me")`
- C. `Button.add("Click Me")`
- D. `Button.new(master)`

8.Consider the following code sequence having five instructions I1 to I5. Each of these instructions has the following format.

OP Ri, Rj, Rk

where operation OP is performed on contents of registers Rj and Rk and the result is stored in register Ri.

I1:ADD R1,R2,R3

I2:MUL R7,R1,R3

I3:SUB R4,R1,R5

I4:ADD R3,R2,R4

I5:MUL R7,R8,R9

Consider the following three statements.

S1: There is anti-dependence between instructions L2 and L5

S2: There is anti-dependence between instructions L2 and L4

S3: Within an instruction pipeline anti-dependence always creates one or more stalls.

Which one of above statements is/are correct?

- A. Only S1 true.
- B. Only S2 true.
- C. Only S1 and S3 are true.
- D. Only S2 and are true.

9.Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first 3 insertions?

- A. $(97 * 97 * 97)/100^3$
- B. $(99 * 98 * 97)/100^3$
- C. $(97 * 96 * 95)/100^3$
- D. $(97 * 96 * 95)/(3! * 100^3)$

10.Consider a hash table with 9 slots. The hash function is $h(k) = k \bmod 9$. The collisions are resolved by chaining. The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. The maximum, minimum, and average chain lengths in the hash table, respectively, are

- A. 3, 0, and 1.
- B. 3, 3, and 3.
- C. 4, 0, and 1.
- D. 3, 0, and 2.

11.Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?

- A. A queue cannot be implemented using this stack.
- B. A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.

- C. A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instructions.
- D. A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each.

12. Suppose a circular queue of capacity $(n - 1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables, respectively. Initially, $\text{REAR} = \text{FRONT} = 0$. The conditions to detect queue full and queue empty are

- A. Full: $(\text{REAR} + 1) \bmod n == \text{FRONT}$
Empty: $\text{REAR} == \text{FRONT}$.
- B. Full: $(\text{REAR} + 1) \bmod n == \text{FRONT}$
Empty: $\text{FRONT} + 1$.
- C. Full: $\text{REAR} == \text{FRONT}$
Empty: $(\text{REAR} + 1) \bmod n == \text{FRONT}$
- D. Full: $(\text{FRONT} + 1) \bmod n == \text{REAR}$
Empty: $\text{REAR} == \text{FRONT}$.

13. Consider the following C program:

```
#include
#define EOF -1
void push (int); /* push the argument on the stack */
int pop (void); /* pop the top of the stack */
void flagError ();
int main () {
    int c, m, n, r;
    while ((c = getchar ()) != EOF) {
        if (isdigit (c) )
            push (c);
        else if ((c == '+') || (c == '*')) {
            m = pop ();
            n = pop ();
            r = (c == '+')? n + m : n*m;
            push (r);
        } else if (c != ' ')
            flagError ();
    }
    printf("%d", pop ());
}
```

What is the output of the program for the following input? 5 2 * 3 3 2 + * +

- A. 15.
- B. 25.
- C. 30.

D. 150.

14. Consider the following ANSI C program.

```
#include <stdio.h>
int main( ) {
int i, j, count;
count = 0;
i = 0;
for (j = -3; j <= 3; j++) {
if ((j >= 0) && (i++))
count = count + j;
}
count = count + i;
printf("%d", count);
return 0;
}
```

Which one of the following options is correct?

- A. The program will compile successfully and output 10 when executed.
- B. The program will compile successfully and output 8 when executed.
- C. The program will compile successfully and output 13 when executed.
- D. The program will not compile successfully.

15. Consider the following array.

23	32	45	69	72	73	89	97
----	----	----	----	----	----	----	----

Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort above array in ascending order?

- A. Insertion sort.
- B. Selection sort.
- C. Quicksort using the last element as pivot.
- D. Merge sort.

16. What is the difference between HTTP and MQTT in IoT?

- A. HTTP is connection-oriented, MQTT is lightweight and publish/subscribe-based
- B. HTTP is for device firmware updates only
- C. MQTT is only for web browsers
- D. HTTP cannot send JSON data

17. Consider the following statements:

- I. The smallest element in a max-heap is always at a leaf node
- II. The second largest element in a max-heap is always a child of the root node
- III. A max-heap can be constructed from a binary search tree in $\Theta(n)$ time
- IV. A binary search tree can be constructed from a max-heap in $\Theta(n)$ time

Which of the above statements are TRUE?

- A. I, II, and III.
- B. I, III, and IV.

C. I, II, and IV.

D. II, III, and IV.

18. Consider the following pseudo code. What is the total number of multiplications to be performed?

D = 2

for i = 1 to n do

 for j = i to n do

 for k = j + 1 to n do

 D = D * 3

A. Half of the product of the 3 consecutive integers.

B. One-third of the product of the 3 consecutive integers.

C. One-sixth of the product of the 3 consecutive integers.

D. None of the above.

19. What does statelessness in REST imply?

A. The server maintains session information for each client

B. Each request from the client must contain all information for processing

C. The client cannot send parameters

D. The server stores user credentials permanently

20. The subset-sum problem is defined as follows: Given a set S of n positive integers and a positive integer W, determine whether there is a subset of S whose elements sum to W. An algorithm Q solves this problem in $O(nW)$ time. Which of the following statements is false?

A. Q solves the subset-sum problem in polynomial time when the input is encoded in unary.

B. Q solves the subset-sum problem in polynomial time when the input is encoded in binary.

C. The subset sum problem belongs to the class NP.

D. The subset sum problem is NP-hard.

Multiple Select Questions (2 Marks)

1. Consider the following recurrence:

$$f(1) = 1;$$

$$f(2n) = 2f(n) - 1, \text{ for } n \geq 1;$$

$$f(2n + 1) = 2f(n) + 1, \text{ for } n \geq 1;$$

Then, which of the following statements is/are TRUE?

A. $f(2^n - 1) = 2^n - 1$.

B. $f(2^n) = 1$.

C. $f(5 \cdot 2^n) = 2^{n+1} + 1$.

D. $f(2^n + 1) = 2^n + 1$.

2. Which of the following is/are valid statement(s) of the spanning tree of a connected undirected graph G (V,E)?

A. Spanning Tree of G(V,E) must contain (V-1) number of edges.

B. Spanning Tree of G(V,E) is minimally connected.

C. Spanning Tree of G(V,E) is maximally cyclic.

D. Spanning Tree of G(V,E) must contain (E-1) number of edges.

3. Which of the following is /are true about the binary relation $s = \phi$ (empty set) on Set $A = \{1, 2, 3\}$ A. Symmetric. B. Irreflexive. C. Reflexive. D. Transitive.
4. Address instructions are supported in: A. AC-based Architecture. B. Register-based Architecture. C. Stack-based Architecture. D. Register-memory based Architecture.
5. Which of the following cache mapping(s) may experience conflict miss? A. Direct Mapping. B. 2-way set associative mapping. C. 4-way set associative mapping. D. Fully associative mapping.
6. A 32-bits wide main memory of size 8GB is formed using A * B-bits DRAM chips. Which of the following is/are true regarding the memory system? A. Chips are arranged in $4 * 4$ if 128M *8-bits chip is used to form the memory. B. 16 number of byte-addressable chips are required each of 512M cells. C. Chips are arranged in $16 * 4$ if 4-bits chip is used to form the memory. D. $5 * 32$ decoder is needed if 64M E. $+ * 4$ if 4-bits chip is used to form the memory.
7. Which of the following can be the value(s) of PC immediately after the fetch of an instruction which is stored on a location 400? A. 400. B. 399. C. 401. D. 402.
8. A processor uses a memory of 4GB capacity in total. Possible address bus sizes are/ are? A. 30. B. 31. C. 32. D. 33.
9. Which of the following statements are true? A. For all inputs, merge sorts runs in $O(n^2)$ time. B. For all inputs, quick sorts runs in $O(n^2)$ time. C. For all inputs, insertion sorts runs in $\Omega(n^2)$ time. D. Every comparison based sorting algorithm runs in $\Omega(1)$ time.
10. Create lexically ordered binary search tree for the following names: raj, ram, joy, sue, ank, ravi, ajay, joe, biku, binod, cat, cow, dog. Which of the followings are true?

- A. Number of leaf nodes in left sub tree of root node are 2.
- B. Number of leaf nodes in right sub tree of root node is 1.
- C. Total number of nodes in left sub tree of root node is 9.
- D. Total number of nodes in right sub tree of root node is 3.

11. Consider the following functions.

```
Void fun(struct node *t)
{
  If(t)
  {
    Fun(t->lchild)
    Printf("%d",t->data);
    Fun(t->rchild);
  }
}
```

Which of the following is / are true?

- A. Function is correct implementation to find in-order traversal of binary tree.
- B. Output print by the function is 4 2 5 1 3 7 6.
- C. Output print by the function is 4 2 5 1 3 6 7.
- D. Output print by the function is 4 2 5 1 7 6 3.

12. A binary search tree is used to locate the number 43. Which of the following probe sequences are possible?

- A. 60 51 13 17 40 43.
- B. 1 2 49 39 59 43.
- C. 11 60 30 48 35 43.
- D. 80 60 50 14 40 43.

13. Which of the following is/ are true for the given tree?

- A. Converse in-order traversal of tree is 5 7 3 1 2 6 4.
- B. Converse pre-order traversal of tree is 1 3 5 7 2 4 6.
- C. Converse post-order traversal of tree is 7 5 3 6 4 2 1.
- D. Reverse of post-order traversal of tree is 1 3 5 7 2 4 6.

14. Which of the following graph is not a topological ordering of the graph?

- A. G B A F D C
- B. G A F B D C.
- C. B D F G A C.
- D. B G D F A C.

15. Which of the following is / are true about below AVL tree?

- A. Converse in-order traversal of AVL tree is 25,20,15,9,8,7,6,3.
- B. In-order traversal of AVL tree is 3,6,7,8,9,15,20,25.
- C. Pre-order traversal of AVL tree is 15,6,3,8,7,9,20,25.
- D. Converse post-order of AVL tree is 3,7,9,8,6,20,25,15.

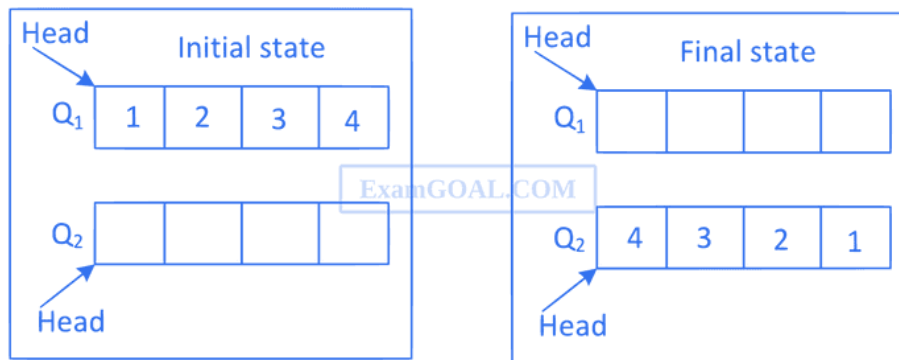
Numerical Answer Type Questions (2 Marks)

1. A block-set associative cache memory consists of 128 blocks divided into four block sets. The main memory consists of 16,384 blocks and each block contains 256 eight bit words. A. How many bits are required for addressing the main memory? B. How many bits are needed to represent the TAG, SET and WORD fields?
2. Consider a direct mapped cache with 8 cache blocks (0-7). If the memory block requests are in the order- 3, 5, 2, 8, 0, 6, 3, 9, 16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82, 17, 24 Which of the following memory blocks will not be in the cache at the end of the sequence? Also, calculate the hit ratio and miss ratio.
3. A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____.
4. A processor has 16 integer registers (R0,R1,...,R15) and 64 floating point registers (F0,F1,...,F63). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating point register operand (1F). The maximum value of N is _____.
5. Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 3rd largest element of the tree is stored at index _____.
6. The postorder traversal of a binary tree is 8,9,6,7,4,5,2,3,1. The inorder traversal of the same tree is 8,6,9,4,7,2,5,1,3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____.
7. Consider the following sequence of operations on an empty stack. <i>push(54); push(52); pop(); push(55); push(62); s = pop();</i> Consider the following sequence of operations on an empty queue.

enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue();

The value of $s + q$ is _____

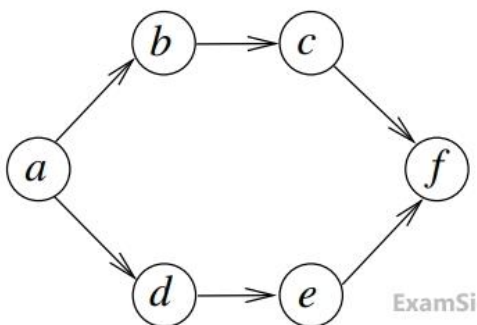
8. Consider the queues Q1 containing four elements and Q2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are Enqueue (Q, element) and Dequeue (Q). The minimum number of Enqueue operations on Q1 required to place the elements of Q1 in Q2 in reverse order (shown as the Final State in the figure) without using any additional storage is _____.



9. Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 23$, and the secondary hash function is $h_2(k) = 1 + (k \bmod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k=90$ is _____.

10. Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n -th vertex in this BFS traversal, then the maximum possible value of n is _____.

11. Consider the following directed graph:



The number of different topological orderings of the vertices of the graph is _____.

12. Let G be a graph with $100!$ Vertices, with each vertex labelled by a distinct permutation of the numbers $1, 2, \dots, 100$. There is an edge between vertices u and v if and only if the label of u can be obtained by swapping two adjacent numbers in the label of v . Let y denote the degree of a vertex in G , and z denote the number of connected

components in G. Then, $y+10z=$ _____.

13. Consider the following C program:

```
#include <stdio.h>
int jumble (int x, int y) {
    x = 2 * x + y ;
    return x ;
}
int main ( ) {
    int x=2, y=5 ;
    y = jumble (y, x) ;
    x = jumble (y, x) ;
    printf ("%d \n", x) ;
    return 0 ;
}
```

The value printed by the program is _____.

14. Consider the following C program:

```
#include <stdio.h>
int counter = 0;
int calc (int a, int b) {
    int c;
    counter++;
    if (b==3) return (a*a*a);
    else {
        c = calc(a, b/3);
        return (c*c*c);
    }
}
int main () {
    calc(4, 81);
    printf ("%d", counter);
}
```

The output of this program is _____.

15. Consider the following ANSI C function:

```
int SimpleFunction (int y[], int n, int x)
{
    int total = y[0], loopIndex;
    for (loopIndex = 1; loopIndex <= n - 1; loopIndex++)
        total = x * total + y[loopIndex];
    return total ;
}
```

Let Z be an array of 10 elements with $Z[i] = 1$, for all i such that $0 \leq i \leq 9$. The value returned by SimpleFunction (Z, 10, 2) is _____

16. Consider the following C functions.

```
int fun1 (int n) {
    static int i = 0;
```

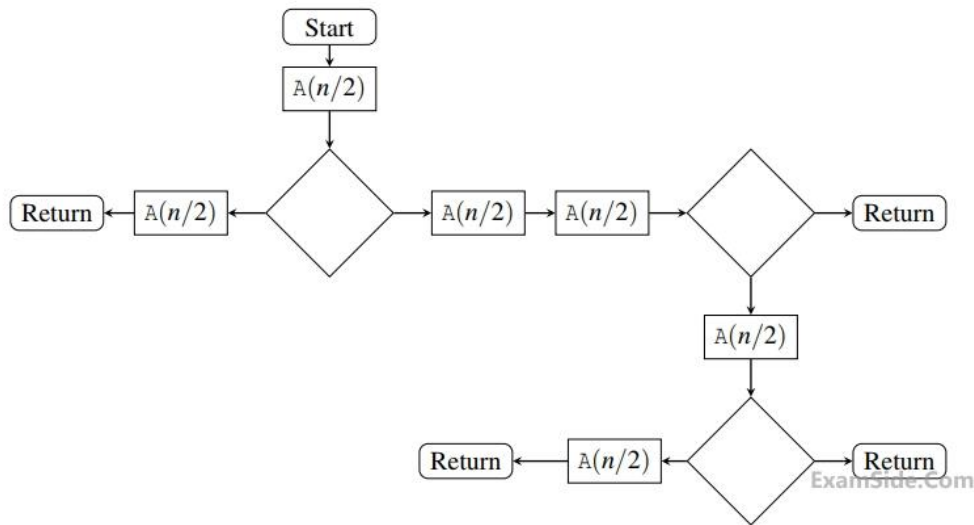

}

}

The return value of fun2 (5) is _____.

17. The given diagram shows the flowchart for a recursive function $A(n)$. Assume that all statements, except for the recursive calls, have $O(1)$ time complexity. If the worst case time complexity of this function is $O(n^\alpha)$, then the least possible value (accurate up to two decimal positions) of α is _____.

Flowchart for Recursive Function $A(n)$



18. An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____.

19. The number of possible min-heaps containing each value from $\{1,2,3,4,5,6,7\}$ exactly once is _____.

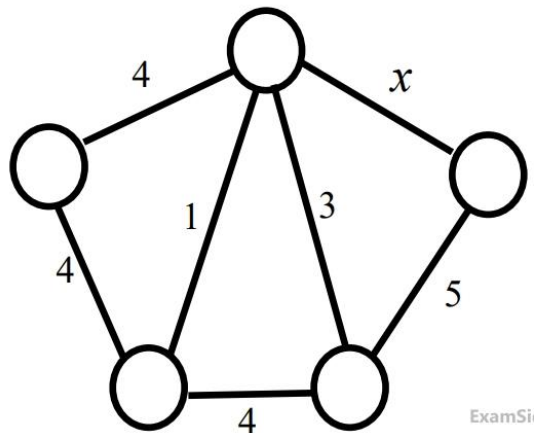
20. A complete binary min-heap is made by including each integer in $[1, 1023]$ exactly once. The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is _____.

21. The minimum number of arithmetic operations required to evaluate the polynomial $P(X) = X^5 + 4X^3 + 6X + 5$ for a given value of X using only one temporary variable is _____.

22. The minimum number of comparisons required to find the minimum and the maximum of 100 numbers is _____.

23. Consider a graph $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_{100}\}$, $E = \{(v_i, v_j) \mid 1 \leq i < j \leq 100\}$, and weight of the edge (v_i, v_j) is $|i - j|$. The weight of the minimum spanning tree of G is _____.

24. Consider the following undirected graph G :



Choose a value for x that will maximize the number of minimum weight spanning trees (MWSTs) of G . The number of MWSTs of G for this value of x is _____.

25. Let A_1, A_2, A_3 , and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$, and 10×5 , respectively. The minimum number of scalar multiplications required to find the product $A_1 A_2 A_3 A_4$ using the basic matrix multiplication method is _____.

Multiple Select Questions (2 Marks)

1. A student wants to build a smart home system that controls lights and monitors temperature using sensors. Which Raspberry Pi suitable for this IoT project? _____

2. A farmer wants to monitor soil moisture in his field and automatically turn on a water pump when the soil becomes dry. Which IoT application using Raspberry Pi can help achieve this?

3. A student installs the operating system Raspbian on a Raspberry Pi to develop an IoT project. After booting, the student wants to use the default graphical desktop to write Python programs

and manage files easily. Which desktop environment is commonly used in Raspbian for this purpose? _____
4. A student is connecting an LED to a Raspberry Pi for an IoT experiment. While writing the Python program, the student wants to use the actual physical pin numbers on the GPIO header (for example, pin 1, pin 2, pin 3 as they appear on the board). Which numbering system should the student use? _____
5. A developer builds an IoT project using a Raspberry Pi that collects temperature data from sensors and sends it to an online dashboard so users can view the data through a web browser. Which protocol should the device use to communicate with the web server? _____
6. A developer wants to create a system that automatically turns on lights when motion is detected using sensors connected to a Raspberry Pi. Which tool is particularly useful for visually connecting these components? _____
7. A developer is building a web-based home automation system on a Raspberry Pi. The system needs to send and receive HTTP requests to control smart devices through web APIs. Which Python library is commonly used to handle HTTP requests in such applications? _____
8. A developer is designing a RESTful API for a smart home system running on a Raspberry Pi. The system needs to add a new device entry to the server database through an HTTP request. Which HTTP method should be used to create a new resource in a RESTful API? _____
9. A developer is creating a graphical user interface for a smart home monitoring system using Tkinter in Python on a Python environment. After designing the interface, the program needs to start the GUI and continuously listen for user events such as button clicks and inputs. Which method is used to run the Tkinter application's event loop? _____
10. An IoT developer is building a web server on a Raspberry Pi to allow mobile apps and other devices to communicate with sensors and actuators over the internet. The developer uses a RESTful API to enable this communication. What is the role of REST APIs in IoT web servers? _____

Section1: Discrete and Engineering Mathematics

Mathematical Logic: propositional and first order logic. Set Theory & Algebra: Sets, relations, junctions, partial orders, lattices and groups. Combinatory: counting, recurrence relations, generating functions. Graph Theory: connectivity, matching, coloring. Probability: Random variables, uniform, normal, exponential, poisson, binomial distribution, mean, Mode, median,

standard deviation, conditional probability and Bayes theorem. Linear Algebra: Matrices, determinates, system of linear equations, Eigen values and Eigen vectors LU decomposition. Calculus: limits, continuity and differentiability, maxima and minima, Mean value theorem, Integration.

Section 2: Digital Logic

Logic Functions, Minimization, Design and Synthesis of Combinational and sequential circuits, number representation and computer arithmetic(fixed and floating point).

Section 3: Computer Organization and Architecture

Machine instructions and addressing modes, ALU and data-path, CPU control design, Memory interface, I/O interface(interrupt and DMA mode),instruction pipelining, cache and main memory, Secondary storage.

Section 4: Programming and Data Structures

Programming in C, Functions, Recursion, Parameter passing, Scope Binding, Abstract data types, Arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs.

Section 5: Algorithms

Analysis, Asymptotic notation, Notions of space and time complexity, worst and average case analysis, Design: Greedy approach, Dynamic programming, Divide and conquer; Tree and graph traversals, connected components, spanning trees, shortest paths, Hashing, sorting, searching, Asymptotic analysis(best, worst, average cases)of time and space, upper and lower bounds, Basic concepts of complexity classes-P, NP, NP-hard, NP-complete.

Section 6: Compiler Design

Lexical Analysis, parsing, Syntax directed translation, Runtime Environments, Intermediate and target code generation, Basic codes of optimization.

Section 7: Operating System

Processes, threads, inter process communication, Concurrency, Synchronization, Deadlock, CPU Scheduling, Memory management and virtual memory, File System, I/O system, Protection and security.

Section 8: Databases

ER-model, relational model(relational algebra, tuple calculus), Database design(integrity constraints, normal forms), Query languages(SQL), File Structures(sequential Files, indexing, B and B+ trees), Transactions and concurrency control.

Section 9: Computer Networks

ISO/OSI stack, LAN technologies(Ethernet and Token ring), Flow and error control techniques, Routing algorithms, congestion control, TCP/UDP and sockets, IP(v4),Application layer protocols(ICMP,DNS,SMTP,POP,FTP,HTTP).Basic concepts of hubs, switches, gateways and routers. Network Security: Basic concepts of public and private key cryptography, digital signature, firewall.

Section 10: Sensors and Actuators

sensors and actuators in IoT, Physical sensors, Smart and MEMS sensors, IoT Applications-Range Sensors, Laser Range Sensors, Common actuators, Actuator control methods, Actuator interfacing with microcontrollers and IoT platforms, CoAP, MQTT.

Section 11: Programming Languages for IoT

Raspberry Pi, Raspbian Linux, Python Programming on Raspberry Pi, Lists and List Methods in Python, RPi.GPIO Library, Functions for GPIO Operations, Tkinter Python Library, Web Server for IoT, Designing a RESTful Web API, LAMP Web Server

Section 12: General Aptitude

Verbal Ability: English, grammar, sentence completion, verbal analysis, word groups, instructions, critical reasoning and verbal deduction, Numerical Ability: Numerical computation, numerical estimation, numerical reasoning and data interpretation.